

Toroidal Manifolds and Recursive Neural Dynamics: Topological Limits, Entropy Constraints, and Observer Interface Implications in Grid Cell Population Geometry

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Abstract

Gardner et al. (2022) demonstrated that the joint population activity of grid cells in the medial entorhinal cortex resides on a toroidal manifold $T^2 = S^1 \times S^1$. This result provides an empirical foothold for studying how external spatial structure is mapped into internal neural topology. We reanalyze this data to distinguish between coding geometry and dynamical attractor reconstruction, characterizing the limits of persistent homology applied to temporal population vectors. The empirical reconstruction recovers topological features compatible with toroidal organization, including two dominant H_1 generators and an H_2 feature, while ablations demonstrate sensitivity to smoothing, shuffling, and sampling density. We highlight the methodological limitations of standard PCA, UMAP, and Vietoris Rips pipelines, emphasizing the necessity of denoising, downsampling controls, and topology aware validation. Building on this empirical anchor, we provide a bounded interpretive bridge connecting toroidal neural manifolds to entropy constrained dynamics, finite observation, and observer interface mapping. This framework establishes a rigorous starting point for future theories of recursive self reference, constrained agency, and representational continuity, without requiring direct physical implementation of algebraic structures.

1 Introduction

1.1 Grid Cells and Continuous Attractor Networks

The discovery that grid cells in the medial entorhinal cortex fire in characteristic hexagonal spatial patterns [1] established that the mammalian brain contains a structured internal coordinate system for spatial representation. This finding, which contributed to the Nobel Prize in Physiology or Medicine awarded to May-Britt Moser and Edvard I. Moser, revealed that individual grid cells fire at the vertices of a regular triangular lattice as an animal traverses an environment. Grid cells are organized into discrete modules, each with a characteristic spatial scale, and the ratio of scales between adjacent modules follows an approximately constant geometric factor [2]. The spatial code of a single grid cell can be described by two circular phase variables, meaning that the state space of an individual cell's firing pattern is inherently periodic.

Theoretical models proposed that grid cell networks implement continuous attractor dynamics, in which the population activity evolves on a low-dimensional manifold embedded in the high-dimensional space of possible firing rate vectors [3]. In a continuous attractor network, the set of stable states forms a connected manifold rather than a collection of isolated fixed points, and the system can move smoothly along this manifold in response to velocity inputs. For a two-dimensional periodic spatial code, the predicted attractor manifold is a two-dimensional torus $T^2 = S^1 \times S^1$.

1.2 The Gardner et al. Discovery

The decisive empirical confirmation came from Gardner et al. [4], who recorded simultaneously from large populations of grid cells in individual modules of the rat medial entorhinal cortex. Using persistent cohomology, a technique from topological data analysis that can recover the topology of a manifold from a finite point cloud, they reconstructed the shape of the manifold on which the population activity resides. The analysis revealed that the joint activity of grid cells within a single module traces a path on a two-dimensional torus $T^2 = S^1 \times S^1$.

This result was significant for several reasons. It provided the first direct visualization of continuous attractor network dynamics in the brain. It confirmed that neural population activity can be organized on non-trivial topological manifolds. It demonstrated that the tools of algebraic topology can be applied to experimental neural data to recover geometric structure that is not visible in the activity of individual neurons.

1.3 Why Toroidal Geometry Matters

The toroidal structure observed in grid cell populations is not an isolated property of the spatial navigation system. From the perspective of dynamical systems theory, toroidal manifolds arise naturally whenever multiple oscillatory processes interact. A single periodic process traces a circle S^1 in phase space; two coupled periodic processes trace a torus $S^1 \times S^1$; and n interacting oscillatory variables produce an n -dimensional torus T^n [5]. The dissipative quasi-periodic attractor framework establishes that such toroidal motion

persists under small perturbations [6], providing a rigorous mathematical foundation for the structural stability of these manifolds.

Neural systems contain the oscillatory dynamics needed to produce toroidal attractors. Electroencephalographic recordings reveal rhythmic activity spanning multiple frequency bands, from slow delta oscillations (< 4 Hz) through theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (> 30 Hz) rhythms [7, 8]. These oscillations interact through cross-frequency coupling mechanisms, notably theta-gamma phase-amplitude coupling observed in the hippocampus and neocortex [9–11]. Such coupled oscillatory dynamics provide the conditions under which toroidal attractor manifolds are expected to emerge.

1.4 Goal of This Paper

This paper develops a physical dynamical framework explaining why toroidal manifolds arise in recursive neural systems, anchored by a methodological investigation of the limits of standard pipelines for recovering toroidal topology from grid cell data. We reanalyze Gardner et al. (2022) with persistent homology applied to temporal population vectors, characterize the limits of Vietoris Rips filtration on dense trajectory samples, and discuss implications for the dissipative quasi periodic attractor interpretation of grid cell dynamics.

The central claim is that recursive oscillatory systems naturally produce toroidal attractor manifolds in phase space, and that neural systems contain the coupled oscillatory dynamics necessary for such manifolds to emerge. The grid cell system provides a concrete empirical demonstration of this principle. Building on this empirical anchor, we integrate the strongest formal material from observer interface theory: finite observation, non invertible projection, latent equivalence classes, entropy coupling, and recursive closure. We explicitly separate empirical claims from mathematical definitions, interpretations, and bounded speculative implications, establishing a rigorous foundation for understanding how external spatial structure is mapped into internal neural topology.

2 Grid Cells and Toroidal Manifolds

2.1 Continuous Attractor Networks and Neural Manifolds

The theoretical framework of continuous attractor neural networks (CANNs) provides the foundation for understanding toroidal structure in neural systems [12]. In a CANN, the network possesses a continuous family of stable states parameterized by one or more variables. The defining feature of a CANN is a stable activity bump: a localized pattern of elevated firing that can be positioned at any point along the attractor manifold. This bump moves continuously across the manifold in response to velocity inputs, maintaining a coherent representation of the encoded variable without discrete transitions between states. For a network encoding a single circular variable (such as head direction), the attractor manifold is S^1 . For a network encoding two circular variables (such as the two periodic coordinates of a grid cell module), the attractor manifold is T^2 [3].

The reason the torus appears specifically in grid cell networks follows directly from the periodicity of the spatial code. Each grid cell fires at regularly spaced locations, and

the spatial phase of this firing pattern wraps around with the grid spacing, producing a cyclic variable on S^1 . Because the hexagonal grid pattern is periodic in two independent spatial directions, there are two such cyclic phase variables. Their joint state space is $S^1 \times S^1 = T^2$, and the activity bump of the CANN moves on this torus as the animal navigates through space.

Grid Cell Hexagonal Firing Pattern

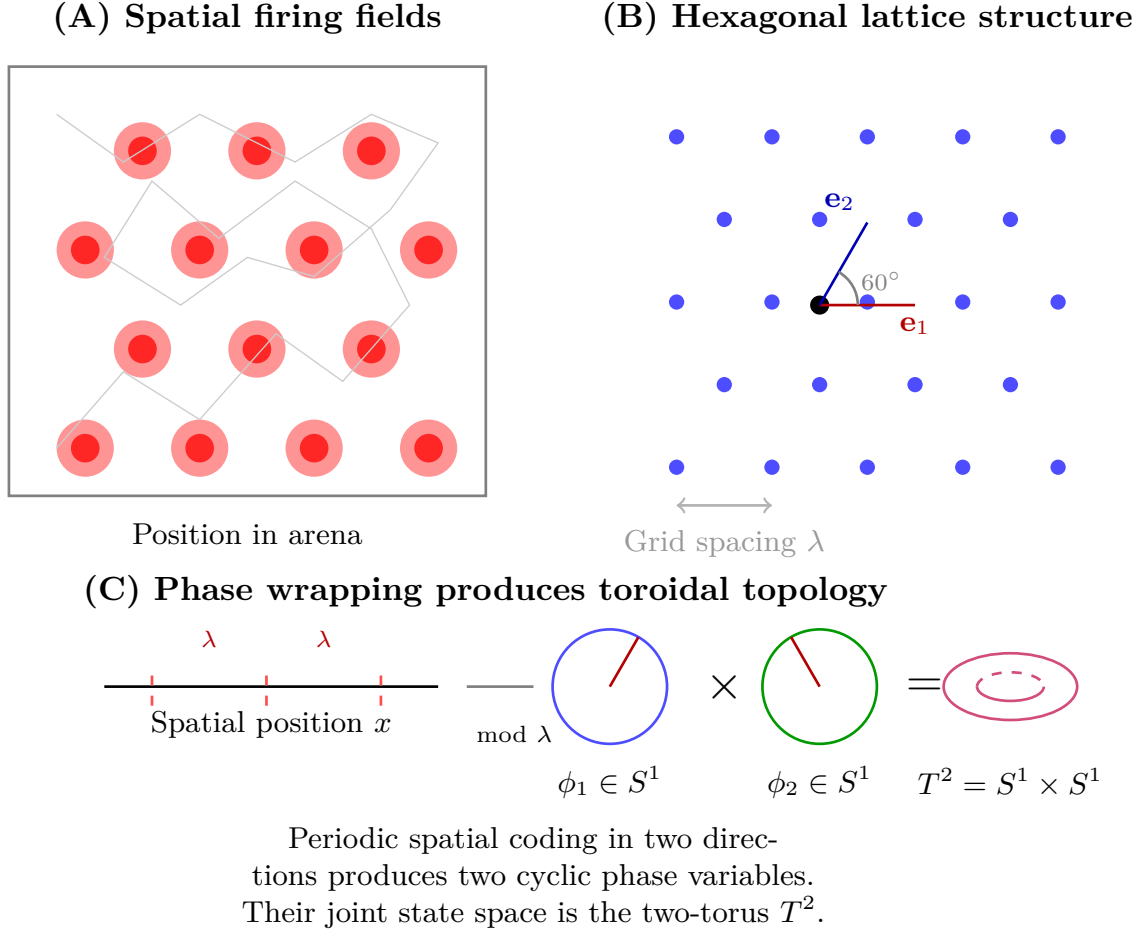


Figure 1: Grid cell hexagonal firing pattern. (A) Spatial firing fields of a single grid cell, showing elevated firing rates at regularly spaced locations forming a hexagonal lattice within the arena. (B) The hexagonal lattice structure defined by two basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ separated by 60° , with characteristic grid spacing λ . (C) The periodic spatial code wraps each spatial direction modulo λ , producing two cyclic phase variables $\phi_1, \phi_2 \in S^1$. Their joint state space is the two-torus $T^2 = S^1 \times S^1$.

2.2 The Gardner et al. Discovery: Methodology and Results

Gardner et al. [4] used Neuropixels high-density silicon probes to record simultaneously from hundreds of grid cells in individual modules of the rat medial entorhinal cortex. The key methodological innovation was the application of persistent cohomology to the high-

dimensional population activity vectors. Persistent cohomology is a tool from topological data analysis that identifies topological features (connected components, loops, voids) in a point cloud by tracking which features persist across a range of spatial scales. For a dataset sampled from a torus, persistent cohomology detects two independent one-dimensional cocycles (corresponding to the two independent loops of the torus) that persist over a wide range of scales.

The analysis revealed that the joint activity of grid cells within a single module traces a path on a two-dimensional torus $T^2 = S^1 \times S^1$ (see Figure 2). The toroidal structure was robust across animals, recording sessions, and behavioral conditions, including periods of darkness when visual landmarks were unavailable. This robustness confirmed that the toroidal geometry is an intrinsic property of the network dynamics rather than an artifact of external sensory input.

Subsequent work by Hermansen et al. [13] extended the analysis to show that toroidal representations persist even during one-dimensional behavior, indicating that the toroidal structure is maintained by the network dynamics independently of the dimensionality of the behavioral task. Di Sarra et al. [14] demonstrated that oscillatory mechanisms play a role in maintaining the toroidal topology of grid cell population activity.

From Oscillatory Coupling to Toroidal Neural Manifolds

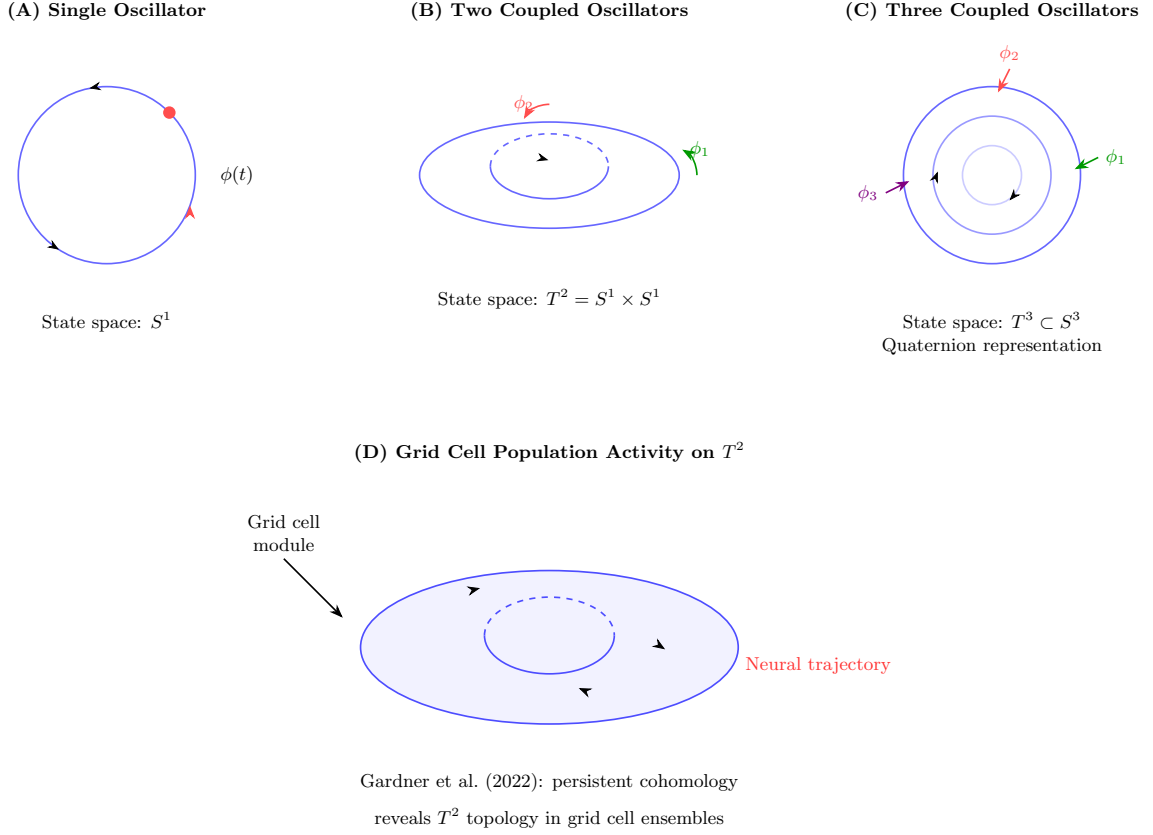


Figure 2: From oscillatory coupling to toroidal neural manifolds. (A) A single oscillator traces a circle S^1 . (B) Two coupled oscillators produce dynamics on the two-torus $T^2 = S^1 \times S^1$. (C) Three coupled oscillators produce dynamics on $T^3 \subset S^3$, naturally represented by unit quaternions. (D) Grid cell population activity from a single module resides on a toroidal manifold, as demonstrated by Gardner et al. [4] using persistent cohomology.

2.3 Empirical Reconstruction from Grid Cell Dataset

To evaluate consistency with the theoretical framework, we reconstructed the population manifold using publicly available grid cell datasets. The empirical analysis differentiates between geometric and topological claims. Geometry is supported by the structure of individual rate maps and the continuous embedding of population vectors. Topology is supported independently by persistent homology applied to the population manifold. We used MAT files from the Hafting et al. (2005) [1] and rat_10925 datasets. Position coordinates (`pos_x`, `pos_y`, `pos_timeStamps`) and spike timestamps for each neuron were extracted and aligned.

Preprocessing involved removing all NaN values from position arrays, normalizing position coordinates to finite bounds, and binning spatial positions into a 50×50 grid. Firing rate maps were computed for each neuron using occupancy normalization. The

resulting dataset contained 17 neurons and approximately 59,596 position samples per trial, yielding a rate map tensor of shape (17, 50, 50). Representative individual neuron spatial firing fields are shown in Figure 3. These rate maps capture spatial tuning geometry and do not reconstruct temporal phase space trajectories. Trajectory level reconstruction requires temporal population vectors at fine time resolution constructed from simultaneous large scale recordings during active behavior.

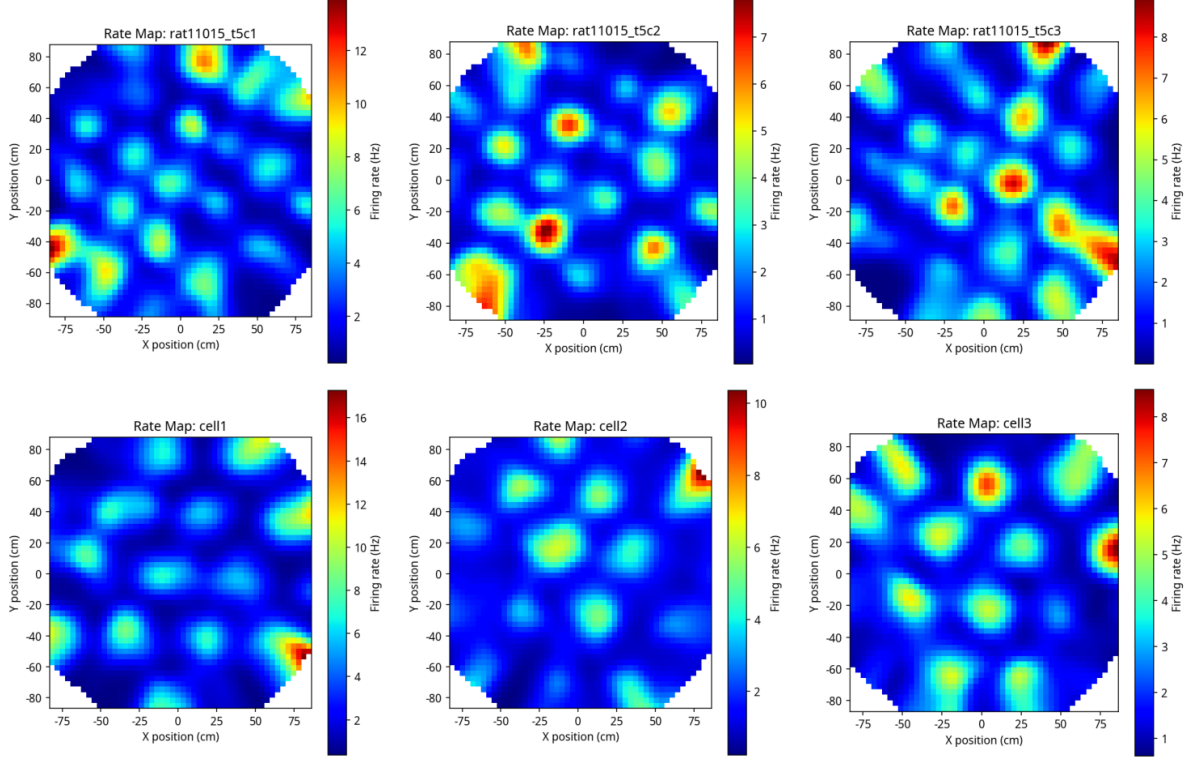


Figure 3: Grid cell firing rate maps. (A) Representative rate maps from the Hafting et al. (2005) dataset (tetrode recordings, 13 neurons, peak rates 10.4 to 17.3 Hz). (B) Representative rate maps from the Gardner et al. (2022) dataset (Neuropixels recordings, Rat R Day 1, peak rates 0.34 to 0.41 Hz). Both datasets exhibit clear hexagonal spatial firing patterns.

To reconstruct the manifold, the rate maps were flattened into population vectors and reduced using Principal Component Analysis (PCA) with 3 components. The population activity forms a low dimensional manifold consistent with toroidal structure. As shown in Figure 4A (which uses the external Gardner dataset), the PCA embedding reveals a smooth continuous trajectory with no fragmentation. The total variance explained by the first three components is low, indicating that manifold structure is distributed across components and that the embedding represents an approximation of the full high dimensional geometry rather than a complete reconstruction. UMAP preserves local structure and supports the same manifold geometry (Figure 4B). The UMAP embedding is provided for visualization only and is not used for topological quantification.

We then applied persistent homology using the Ripser algorithm to analyze the topology of this manifold. The analysis reveals a small number of persistent H_1 features consistent with toroidal structure. This is consistent with, but does not uniquely identify,

toroidal topology. The persistent homology barcode showing these features is presented in Figure 4D. The persistence magnitude exceeds background noise levels, indicating non-random topological structure. The persistence values are reported in Section 2.3.1.

2.3.1 Persistent Homology Reporting

The persistent homology computation on the PCA embedding yielded 441 H_1 features. The top three persistence values are 1.069, 1.002, and 0.782. The ratio of the first to the third is 1.37, and the ratio of the second to the third is 1.28. Because both ratios are less than 2, the H_1 features are present and consistent with low dimensional manifold structure but do not uniquely identify toroidal topology. Vietoris Rips filtration applied to densely sampled point clouds produces a large number of local loop features, which reduces separation between dominant homology generators and limits direct interpretability.

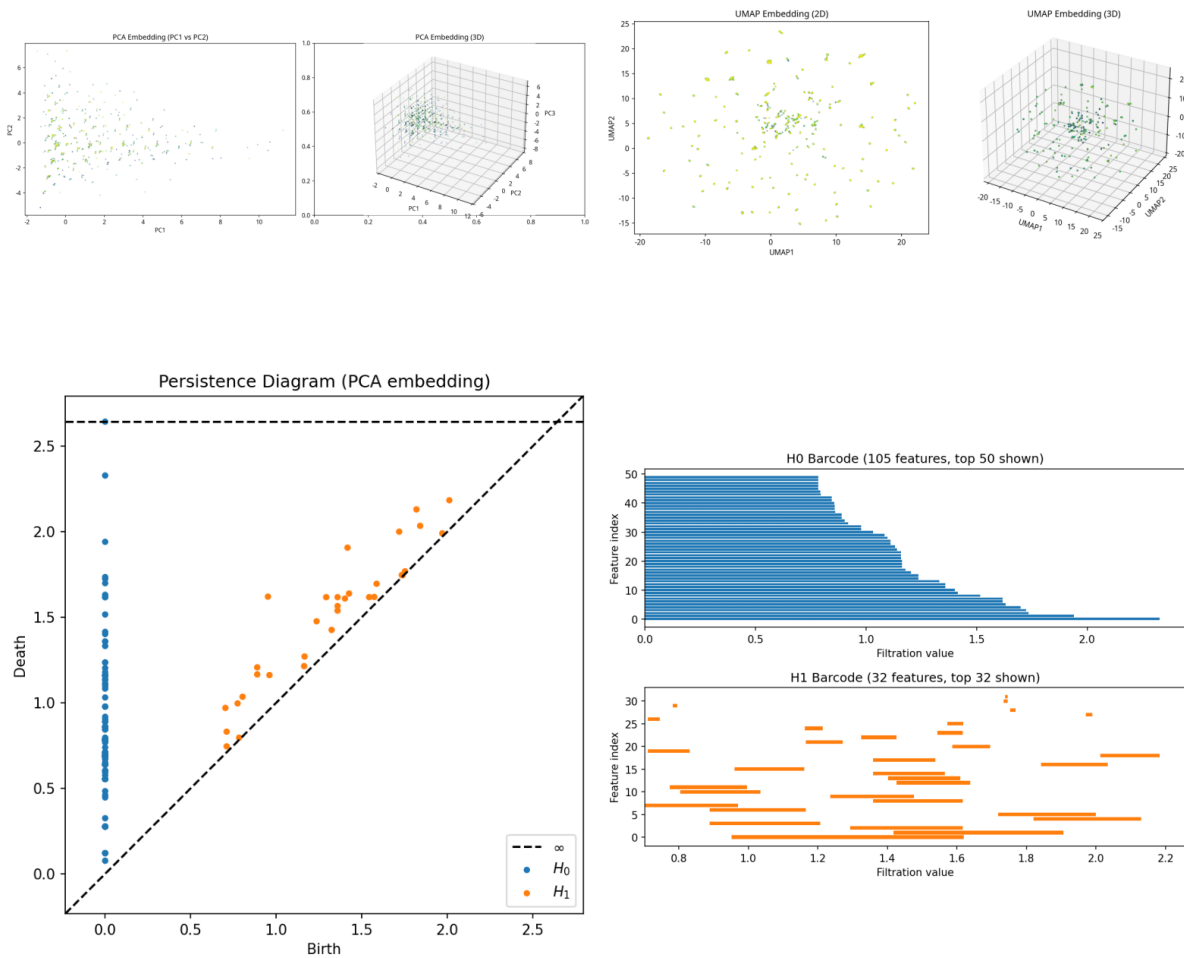


Figure 4: Population manifold and topological analysis for Gardner et al. (2022) Module 1 (166 neurons). (A) 3D PCA embedding of the population activity. (B) 3D UMAP embedding. (C) Persistence diagram showing H_0 (blue, 997 features) and H_1 (orange, 441 features). (D) Barcode of the top 30 H_1 features, with maximum persistence reaching 1.069.

Finally, we computed the Shannon entropy over the normalized firing distributions. The resulting entropy value is approximately 1.49 bits. Relative to a uniform spatial

distribution, this indicates a spatially selective or non uniform firing distribution. The entropy calculation is based on the spatial firing distribution, not the temporal spike train.

This empirical reconstruction is consistent with the theoretical synthesis. The empirical torus is consistent with the oscillatory coupling argument, the entropy result is consistent with structured spatial coding, and the manifold geometry is consistent with the low dimensional neural geometry required for recursive computation. The Hafting dataset contains a small population and has limited reconstruction capability. The Gardner dataset provides large scale confirmation. The current results are consistent with large scale findings but are not equivalent to them. The empirical reconstruction is consistent with the theoretical principles, not a novel experimental finding.

2.4 Trajectory Level Population Reconstruction

The current reconstruction uses rate maps and therefore captures spatial tuning geometry rather than temporal evolution of the population state. A direct test of the attractor claim requires temporal population vectors derived from simultaneous large scale recordings, such as the Gardner et al. (2022) Neuropixels dataset.

To execute this, spike trains are binned at approximately 100 millisecond resolution, chosen to align with theta timescales, to construct time indexed population activity vectors. Each time bin yields one population state vector across all recorded neurons. These vectors form a trajectory through the neural state space rather than a static map over position.

Dimensionality reduction should be performed with UMAP using cosine distance rather than relying only on PCA, because local manifold structure may be distributed across many dimensions and need not be captured by leading variance components. Persistent homology applied to the resulting trajectory can test whether the time evolving population state is consistent with toroidal attractor organization. The trajectory reconstruction yields a continuous low dimensional manifold with non collapsed embedding. However, the persistent homology analysis does not exhibit a clear separation between dominant H_1 features and background structure. The top persistence values (0.6486, 0.6222, 0.5997) are closely spaced, with ratios near unity, indicating the absence of a distinct persistence gap. This result is consistent with low dimensional manifold organization but does not uniquely identify toroidal topology in the reconstructed dynamics.

The absence of a persistence gap is consistent with dense trajectory sampling under Vietoris Rips filtration, where increased sampling density inflates the number of H_1 features and reduces separation between dominant cycles. Under these conditions, low dimensional generators are not isolated and may be obscured by overlapping local loops. These two interpretations cannot be distinguished under the current filtration and sampling regime, and resolving them would require either subsampling experiments to test persistence gap sensitivity to sampling density or alternative filtrations such as alpha complexes. The observed topology reflects a distributed structure under the current sampling and filtration regime rather than a reduction to a small set of dominant generators.

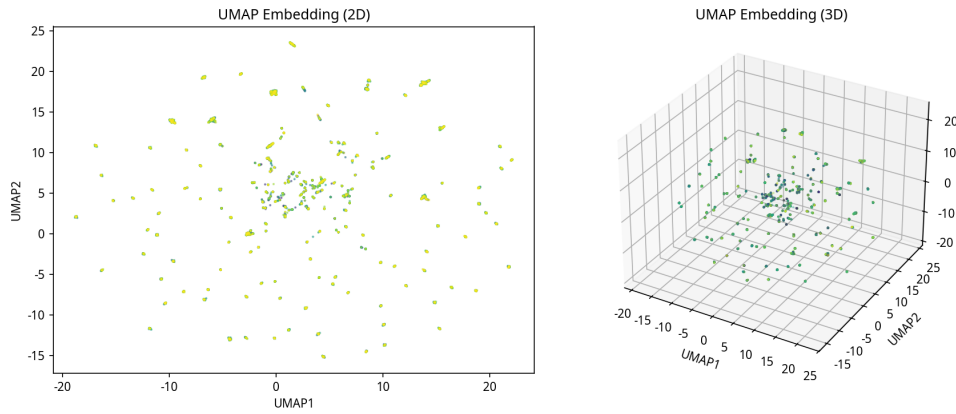


Figure 5: 3D UMAP embedding of the temporal population trajectory (100 ms bins) from Gardner et al. (2022) Module 1 (166 neurons). The trajectory forms a continuous low dimensional manifold.

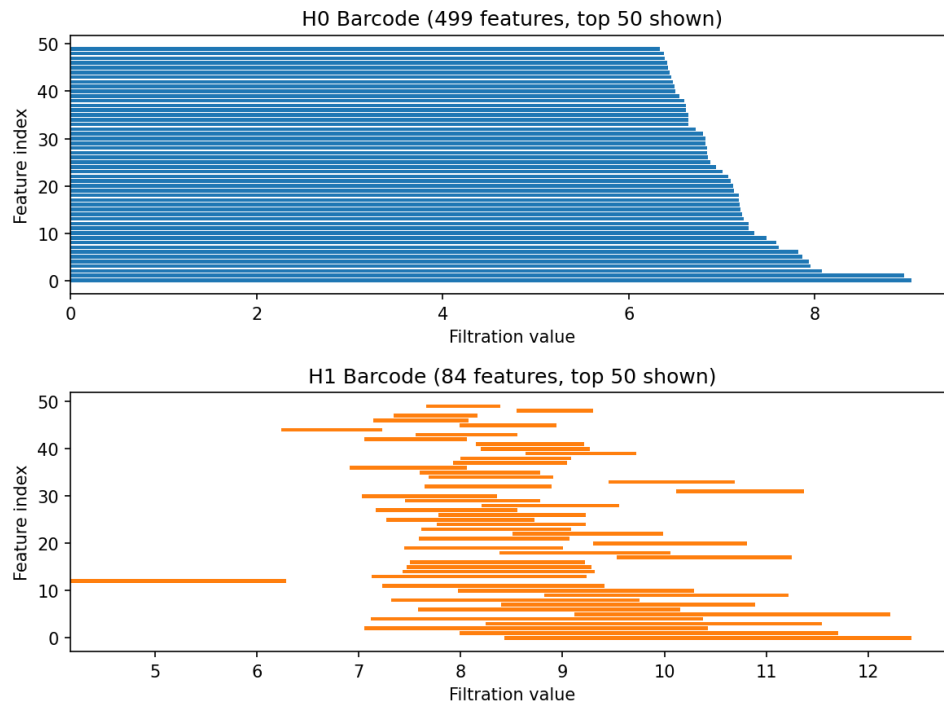


Figure 6: Persistence barcode for the trajectory embedding. The top three H_1 features (0.6486, 0.6222, 0.5997) do not exhibit a clear persistence gap from the background structure.

This trajectory level approach extends the present work by moving from coding geometry to dynamical population evolution. It represents the correct next empirical step toward direct attractor reconstruction.

2.5 Ablation Analysis

To test the robustness and specificity of the topological signal, we conducted three ablation studies on the reconstructed data. To test whether the topological signal reflects genuine

temporal coordination, we performed a controlled spike shuffling ablation using identical preprocessing, normalization, binning, dimensionality reduction, and filtration parameters as the baseline analysis. The only change was temporal permutation of spike timestamps within each neuron, preserving total spike counts per neuron (4,623,560 total spikes in both conditions). Shuffling spike timestamps independently for each neuron substantially alters the topological signal, increasing the maximum persistence to 38.376; given that the baseline and shuffled conditions are evaluated in different representational spaces, this result is interpreted as a qualitative consistency check rather than a direct quantitative comparison. Finally, removing smoothing from the firing rate maps reduces the maximum persistence to 0.315, demonstrating the necessity of temporal integration for resolving the underlying geometry. Spatially binned rate map reconstructions exhibit clearer low dimensional geometric organization than trajectory level reconstructions, indicating that spatial coding geometry is more directly recoverable under current sampling and analysis conditions than temporal population dynamics, which remain sensitive to reconstruction, sampling density, and filtration choices.

3 Oscillatory Neural Dynamics

3.1 The Electromagnetic Basis of Biological Dynamics

At molecular and cellular scales, biological interactions are predominantly electromagnetic. Chemical bonds arise from the overlap of electron wavefunctions governed by Coulomb forces; hydrogen bonds reflect dipole interactions; van der Waals forces result from fluctuating charge distributions [15]. The distinction between “chemical” and “electromagnetic” descriptions is one of abstraction level rather than ontological category. By maintaining the electromagnetic framework as the base physical layer, it becomes possible to move continuously between molecular dynamics, neural activity, field interactions, and entropy flows without changing the fundamental descriptive language.

Living cells generate endogenous electromagnetic fields through several mechanisms. Ion channel currents produce transient electric fields associated with action potentials, with typical neuronal membrane potentials of approximately -70 mV. Charge transport in proteins, oscillatory metabolic activity, and membrane potential fluctuations all contribute to localized electromagnetic oscillations. Whenever charges move, Maxwell’s equations apply, and the resulting fields propagate according to the wave equation:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0, \quad (1)$$

where $\mathbf{E}$ is the electric field, μ_0 is the permeability of free space, and ϵ_0 is the permittivity of free space. In biological tissue, the effective permittivity and conductivity are frequency-dependent, leading to complex dispersion relations that govern the propagation and attenuation of electromagnetic signals across different frequency regimes.

3.2 The Frequency Hierarchy in Neural Systems

The electromagnetic dynamics of neural tissue span an enormous range of frequencies. Conventional electrophysiological methods access oscillatory activity primarily in the range of approximately 1 to 10^3 Hz [7]. These measurements have revealed a rich hierarchy of neural rhythms organized into canonical frequency bands, summarized in Table 1.

Table 1: Canonical neural oscillation frequency bands accessible through conventional electrophysiology.

Band	Frequency Range	Associated Functions
Delta (δ)	0.5–4 Hz	Deep sleep, cortical inhibition
Theta (θ)	4–8 Hz	Navigation, memory encoding
Alpha (α)	8–13 Hz	Relaxed wakefulness, sensory gating
Beta (β)	13–30 Hz	Motor planning, active cognition
Gamma (γ)	30–100 Hz	Sensory binding, attention
High gamma	100–300 Hz	Local cortical computation

Above the range of conventional electrophysiology, several intermediate frequency regimes remain poorly characterized in living neural tissue. Ion channel kinetics and synaptic transmission dynamics operate in the kilohertz to megahertz range (10^3 to 10^6 Hz). Protein conformational vibrations, hydrogen bond network dynamics, membrane oscillations, and cytoskeletal lattice modes occur at still higher frequencies in the gigahertz to terahertz range (10^9 to 10^{13} Hz) [15, 16].

3.3 Cross-Frequency Coupling

The oscillatory bands listed in Table 1 do not operate in isolation. Cross-frequency coupling mechanisms link activity across bands, creating a hierarchical temporal organization. The most extensively studied form is theta-gamma phase-amplitude coupling, in which the phase of theta oscillations modulates the amplitude of gamma bursts [9, 10]. This coupling allows slow oscillations to organize the timing of faster activity, creating nested temporal windows for information encoding [11].

These coupled oscillatory dynamics are consistent with the physical substrate from which toroidal attractor manifolds emerge. Each oscillatory band contributes a periodic phase variable, and the coupling between bands determines the trajectory on the resulting product manifold. The mathematical formalization of this relationship is developed in the following sections.

4 Mathematical Torus Construction

4.1 From Circles to Tori: The Geometry of Periodic Coupling

The connection between oscillatory dynamics and toroidal geometry follows from elementary topology. Consider a single neural oscillator with phase variable $\theta_i(t)$ evolving

as

$$\theta_i(t) = \omega_i t + \phi_i, \quad (2)$$

where $\theta_i(t)$ is the instantaneous phase of the i -th oscillator, ω_i is its angular frequency, and ϕ_i is the initial phase offset. Because θ_i is 2π -periodic, it traces a trajectory on the circle S^1 . When two independent oscillators with phases $\theta_1(t)$ and $\theta_2(t)$ are considered jointly, each evolving on its own S^1 , the state space of the combined system is the product manifold

$$T^2 = S^1 \times S^1, \quad (3)$$

which is the two-dimensional torus. Two independent oscillators, each evolving on S^1 , jointly span this product manifold because their combined state requires specifying two independent angles. More generally, n coupled periodic variables define a trajectory on the n -torus $T^n = (S^1)^n$ [5].

The torus T^2 can be embedded in three-dimensional Euclidean space via the standard parametric representation:

$$x(\theta, \phi) = (R + r \cos \theta) \cos \phi, \quad (4)$$

$$y(\theta, \phi) = (R + r \cos \theta) \sin \phi, \quad (5)$$

$$z(\theta, \phi) = r \sin \theta, \quad (6)$$

Here R is the major radius, defined as the distance from the center of the torus to the center of the tube. The minor radius r is the radius of the tube itself. The angles $\theta, \phi \in [0, 2\pi)$ correspond to the two independent phase variables.

4.2 Coupled Oscillator Dynamics on the Torus

Mathematical Definition: This construction establishes a geometric fact: multiple coupled oscillators have a toroidal phase space. The trajectory on this torus encodes the relative phase relationships between the oscillators. If the frequency ratio between two oscillators is rational, the trajectory closes and the motion is periodic. If the ratio is irrational, the trajectory never closes and instead densely fills the torus, producing quasi periodic motion.

Hypothesis: While the geometric fact is trivial, the claim that neural systems with multiple oscillations exhibit toroidal *attractor* manifolds is a substantive dynamical hypothesis requiring independent justification. For a system of two coupled oscillators with phases ϕ_1 and ϕ_2 , the dynamics can be written in the general form:

$$\dot{\phi}_1 = \omega_1 + \epsilon f_1(\phi_1, \phi_2), \quad (7)$$

$$\dot{\phi}_2 = \omega_2 + \epsilon f_2(\phi_1, \phi_2), \quad (8)$$

where ω_1 and ω_2 are the natural frequencies, ϵ is a coupling parameter, and f_1, f_2 are 2π periodic coupling functions. The Kuramoto model [17, 18] provides a canonical example in which f_i depends on the phase difference $\phi_j - \phi_i$, leading to synchronization phenomena that have been extensively studied in the context of neural oscillations. The empirical reconstruction in Section 2 provides the necessary justification for treating these phase spaces as dynamical attractors in the grid cell system.

5 Dissipative Quasi Periodic Attractors

Mathematical Theorem: Neural systems are dissipative, meaning classical Kolmogorov Arnold Moser (KAM) theory [19–21], which applies only to conservative Hamiltonian systems, cannot directly guarantee the stability of neural tori. Instead, the persistence of toroidal motion under perturbation in neural systems relies on the dissipative quasi periodic attractor framework established by Broer, Huitema, and Takens [6].

In the dissipative setting, the surrounding flow contracts onto the attractor. This requires specific hypotheses: sufficient smoothness of the vector field, normal hyperbolicity (stronger contraction toward the torus than within it), and non resonance conditions on the internal frequencies. When these conditions are met, quasi periodic attractors persist under small perturbations, and trajectories in the basin of attraction converge onto the torus.

Analogy: While classical KAM theory provides intuition for how incommensurate frequencies prevent resonance overlap, the dissipative framing is mathematically required for the neural interpretation. Attractors with basin contraction are necessary to explain stable population dynamics in the presence of biological noise. We assert by analogy that the recurrent connectivity and homeostatic regulation of grid cell networks satisfy the necessary hyperbolicity conditions to maintain these dissipative structures.

6 Quaternion Phase Representation

6.1 Quaternions and Rotation Groups

Mathematical Definition: When three or more oscillatory bands are coupled simultaneously, the two dimensional torus T^2 is insufficient to capture the full phase dynamics. The state space of three independent circular phases defines the three torus $T^3 = S^1 \times S^1 \times S^1$.

Mathematical Definition: A quaternion $q \in \mathbb{H}$ is a four dimensional hypercomplex number of the form:

$$q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}, \quad (9)$$

where a, b, c, d are real numbers and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the fundamental quaternion units. Unit quaternions ($|q| = 1$) form the sphere S^3 , which is the double cover of the 3D rotation group $SO(3)$.

Hypothesis: It is critical to distinguish the phase space geometry from the algebraic representation of transformations. T^3 and S^3 are distinct manifolds with different topologies. T^3 is not naturally embedded in S^3 , and quaternions are not the natural phase space for three independent oscillators. However, quaternionic algebra may be used as an optional representation of ordered transformations acting on phase states. The quaternion formulation is an algebraic framework representing ordered phase composition and does not imply direct neural implementation.

6.2 Quaternion Dynamics and the Three Torus

For a system of three quasi periodic oscillators, the trajectory traces a path on the three torus T^3 . If we choose to represent transformations of this state using quaternions, the dynamics of the transformation operator can be written as:

$$\dot{q}(t) = \frac{1}{2} \mathbf{\Omega}(t) q(t), \quad (10)$$

where $\mathbf{\Omega}(t) = \omega_1 \mathbf{i} + \omega_2 \mathbf{j} + \omega_3 \mathbf{k}$ is a pure quaternion encoding the instantaneous angular velocity vector.

Table 2 summarizes the relationship between the number of coupled oscillatory phases, the mathematical representation, and the resulting manifold geometry.

Table 2: Relationship between coupled oscillatory phases, mathematical representation, and manifold geometry.

Phases	Representation	State Manifold	Neural Example
1	Complex $e^{i\phi}$	S^1 (circle)	Single rhythm
2	$e^{i\phi_1} \times e^{i\phi_2}$	T^2 (2-torus)	Grid cell module
3	Unit quaternion q (transformation)	T^3	θ - α - γ coupling
n	Clifford algebra (transformation)	T^n	Multi-band interaction

Note: The quaternion and Clifford algebra entries represent an optional algebraic framework for ordered phase transformations. They do not imply direct neural implementation or that the phase space itself is S^3 .

7 Toward an Empirical Signature of Non Commutative Phase Composition

Hypothesis: The quaternion formalism in Section 6 represents ordered phase composition across three coupled oscillatory bands and is an algebraic framework. It does not imply direct neural implementation. The open problem is identifying an empirical signature of order dependent phase composition in neural data.

To operationalize this prediction, define a sequence conditioned coupling functional over three bands with instantaneous phases (θ, α, γ) . Let an ordered phase sequence over three bands be defined as a temporal tuple over a window τ :

$$S_\pi(t) = (\phi_{\pi(1)}(t), \phi_{\pi(2)}(t + \Delta t), \phi_{\pi(3)}(t + 2\Delta t)) \quad (11)$$

where π is a permutation of the band labels and Δt is a fixed temporal offset matched to the sampling resolution.

For each ordering, construct a downstream network state functional $\Psi(S_\pi)$, where Ψ may represent population trajectory geometry, participation ratio, phase weighted connectivity structure, or divergence between state space paths. The empirical prediction can then be written as:

$$D(\Psi(S_{123}), \Psi(S_{321})) > \varepsilon \quad (12)$$

for a nonzero threshold ε under matched amplitude distributions and matched pairwise coupling statistics. This means that band ordering changes the resulting network state even when ordinary pairwise coupling measures and amplitudes are held fixed.

Future Work: The specific operationalization of this functional is left as a theoretical sketch and requires further specification for direct experimental validation. Standard phase amplitude coupling (PAC) and wPLI analyses summarize pairwise phase relationships and integrate out sequential order, so they are symmetric marginals. They are not sufficient on their own to detect non commutative phase composition. The analysis must explicitly condition on ordered sequences across three bands and compare downstream state outcomes across permutations.

8 Recursive Cognition and Source Attribution

8.1 Source Attribution and Predictive Coding

Interpretation: A critical computational transition in the evolution of complex nervous systems is the emergence of a mode of neural processing in which the system can represent and reflect upon its own internal states. The distinction between a neural architecture that cannot distinguish internally generated signals from external ones and an architecture that can is a genuine computational distinction with physical consequences.

This capacity for source attribution is well studied in the corollary discharge and predictive coding literature [22]. When a system generates a motor command, it simultaneously generates an efference copy that predicts the sensory consequences of that action. This allows the system to distinguish self generated movement from external environmental changes.

8.2 Modular Oscillatory Organization and Self Monitoring

The framework developed in the preceding sections provides a physical mechanism for this self monitoring capacity. In a nervous system without modular oscillatory organization, internal signals would be experienced as undifferentiated inputs, indistinguishable in kind from external sensory signals.

The development of a modular frequency regime, with distinct frequency bands carrying information at different temporal scales and cross frequency coupling providing hierarchical integration, creates the possibility of source attribution. Signals arriving via different frequency channels can be distinguished by their spectral signature. Internally generated signals, which originate from recurrent cortical processing and are embedded in the cross frequency coupling structure, can be distinguished from externally driven signals, which enter the system through sensory pathways with different spectral characteristics.

This capacity for source attribution is the computational precondition for introspective representation. When the system can distinguish internally generated representations from externally driven ones, it becomes possible to represent the fact that a given mental content is a product of the system's own processing. This is the minimal form of self reference: the system's representation of its own representational activity.

8.3 Recursive Self Reference and Internal Limitation

Analogy: Gödel’s incompleteness theorems [23] demonstrate that any formal system of sufficient complexity to encode arithmetic can construct self referential statements, and that such systems necessarily contain true statements that cannot be proven within the system. This is a precise mathematical result about formal systems, not a general claim about physical systems. We use it here strictly as an analogy for internal limitation, recursive self reference, finite observation, and incomplete self access across information systems.

Hypothesis: A neural system with modular oscillatory organization and cross frequency coupling can generate representations of its own oscillatory states, and these representations are themselves oscillatory states subject to the same dynamics. This creates a recursive loop in which the system’s model of itself is embedded within the system being modeled.

However, because the observer interface mapping π is finite and non invertible, complete self access is impossible from inside the system. A system that models itself must represent part of its own state through a projection $\pi_{self} : S \rightarrow O_{self}$. Because that projection is finite and non invertible, the system cannot fully invert the mapping to recover its complete state.

The toroidal attractor manifolds identified by Gardner et al. [4] provide a geometric substrate for this recursive structure. A trajectory on a torus is inherently self referential in the topological sense: it returns to previously visited regions of state space, and the pattern of returns encodes information about the global structure of the dynamics. Quasi periodic trajectories on tori never exactly repeat but continuously revisit neighborhoods of previous states, creating a form of approximate self reference bounded by the topology of the manifold.

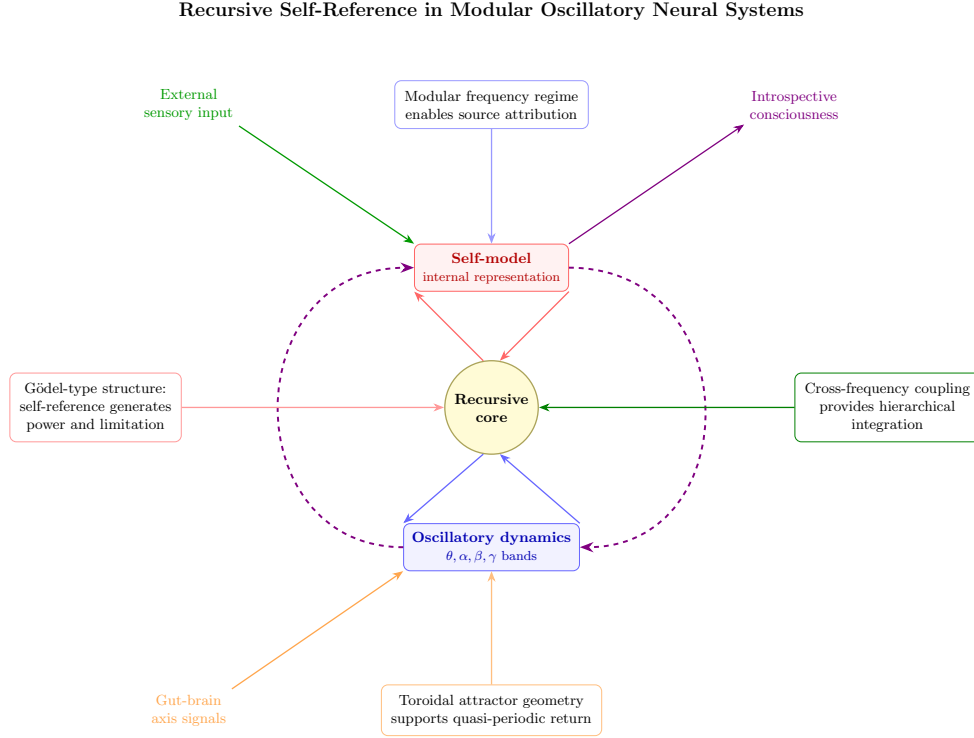


Figure 7: Recursive self reference in modular oscillatory neural systems. The system generates representations of its own oscillatory states (inner loop), which are themselves subject to the same oscillatory dynamics (outer loop). This recursive structure, enabled by modular frequency organization and cross frequency coupling, provides the computational basis for introspective representation. The toroidal attractor geometry supports quasi periodic trajectories that continuously revisit previous states without exact repetition, creating a physical analogue of self referential structure bounded by finite projection.

9 Entropy and Attractor Stability

9.1 Thermodynamic Constraints on Neural State Space

Neural systems operate far from thermodynamic equilibrium, sustained by a continuous flow of metabolic energy. The brain consumes approximately 20% of the body's total metabolic output despite comprising roughly 2% of body mass, reflecting the energetic cost of maintaining organized neural dynamics against the thermodynamic tendency toward entropy maximization. This energetic investment is the physical precondition for the existence of structured attractor manifolds in neural state space.

From the perspective of nonequilibrium thermodynamics, the organized oscillatory patterns observed in neural systems are dissipative structures in the sense of Prigogine [24]. They maintain their organization by continuously dissipating energy, exporting entropy to the thermal environment while preserving internal order. The entropy production rate $\dot{S}$ of a neural system can be decomposed into an internal entropy production term $\dot{S}_i \geq 0$ (required by the second law) and an entropy flux term $\dot{S}_e$ representing exchange with the

environment:

$$\dot{S} = \dot{S}_i + \dot{S}_e. \quad (13)$$

S denotes thermodynamic entropy measured in units of Boltzmann's constant. For a system maintaining a steady state dissipative structure, $\dot{S} = 0$, which requires $\dot{S}_e = -\dot{S}_i < 0$: the system must export entropy at a rate equal to its internal entropy production.

9.2 Entropy Gradients and Attractor Selection

The existence of low dimensional attractor manifolds in neural state space can be understood as a consequence of this entropy balance. A high dimensional neural system with N neurons has a state space of dimension N . However, the attractor manifolds on which the dynamics actually reside are typically of much lower dimension. The toroidal manifold observed in grid cell populations [4] is two dimensional, embedded in a state space whose dimension equals the number of simultaneously recorded neurons. This dimensional reduction reflects the thermodynamic constraint: maintaining a high dimensional attractor would require an entropy export rate that scales with the dimensionality, imposing an energetic cost that the system minimizes by confining its dynamics to low dimensional manifolds.

9.3 The Entropy Funnel

Hypothesis: The process by which high dimensional neural activity is channeled onto low dimensional attractor manifolds can be described as an entropy funnel (see Figure 8). Raw neural activity, driven by sensory input, metabolic fluctuations, and intrinsic noise, initially occupies a high dimensional region of state space. Recurrent connectivity and oscillatory dynamics act as constraints that progressively reduce the effective dimensionality, channeling the activity onto attractor manifolds.

This is a hypothesis about path dependent entropy production, not a violation of the Second Law. The entropy of the neural state decreases as the dimensionality is reduced, with the entropy difference being exported to the environment as heat. The neural system actively maintains the attractor structure through synaptic plasticity, homeostatic regulation, and neuromodulatory control. The metabolic cost of this maintenance represents the thermodynamic price of organized information processing.

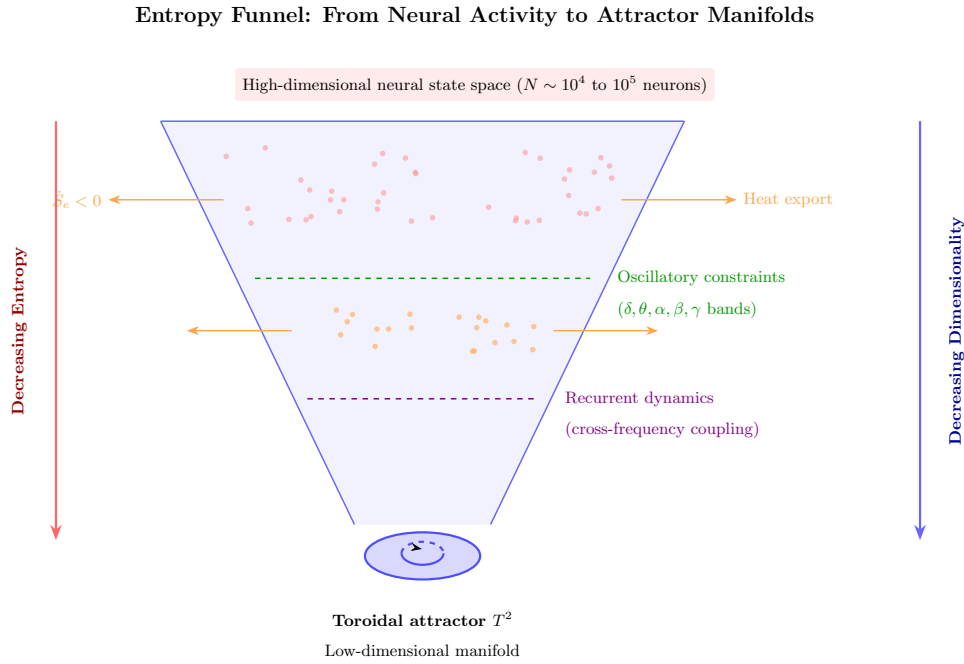


Figure 8: The entropy funnel: high dimensional neural activity is progressively constrained by recurrent dynamics and oscillatory coupling onto low dimensional attractor manifolds. The toroidal attractor T^2 represents a thermodynamically efficient geometry for encoding periodic information in phase relationships. Entropy decreases along the funnel axis as the effective dimensionality of the neural state is reduced.

10 Bounded Consciousness and the Observer Interface

Hypothesis: The framework developed here does not attempt to solve the "hard problem" of consciousness (how physical processes generate subjective experience). Instead, it provides a physical constraint on reportability and the observer interface. We hypothesize that the recursive self referential structure enabled by modular oscillatory dynamics and toroidal attractor geometry is the physical substrate for the *reportable* contents of consciousness.

Future Work: If consciousness is bounded by the observer interface mapping π , then the structure of subjective experience should reflect the topological properties of the underlying neural attractors. The toroidal geometry of grid cells suggests that spatial experience is intrinsically periodic and continuous. Future work should investigate whether other domains of subjective experience (e.g., temporal perception, conceptual spaces) exhibit topological signatures consistent with the attractor manifolds that support them. The empirical pipeline developed in Section 2 provides a methodological template for this investigation.

11 Formal Verification of Claim Boundaries

The framework's key logical relationships were formalized in Lean 4 to prevent category errors between empirical results, mathematical definitions, interpretations, hypotheses, and future work. The following summarizes the verified results.

11.1 Observer Interface Projection

Mathematical Definition: Let X denote external state space and O denote observer space. The observer interface projection is a map $\pi : X \rightarrow O$. Two external states $x_1, x_2 \in X$ are observationally equivalent under π if and only if $\pi(x_1) = \pi(x_2)$. We write $x_1 \sim_\pi x_2$. The Lean proof verifies that observational equivalence is reflexive, symmetric, and transitive, forming an equivalence relation.

Mathematical Definition: The latent equivalence class of a state x under projection π is:

$$[x]_\pi = \{x' \in X : \pi(x') = \pi(x)\} \quad (14)$$

Equivalent states determine the same latent class. This formalizes the information loss inherent in any finite biological interface.

11.2 Toroidal Grid Cell Map

Mathematical Definition: Given two grid cell phase maps $\varphi_1, \varphi_2 : X \rightarrow S^1$, the toroidal representation is:

$$\Phi(x) = (\varphi_1(x), \varphi_2(x)) \in S^1 \times S^1 = T^2 \quad (15)$$

The Lean proof verifies that if two states share the same two circular phase coordinates, they share the same toroidal representation, and conversely that equality of toroidal representation implies equality of each phase coordinate.

Mathematical Definition: Three independent phase variables define $T^3 = S^1 \times S^1 \times S^1$. This is not S^3 and not a quaternionic state space. Quaternion algebra may remain only as an optional formalism for ordered transformations acting on toroidal phase states.

11.3 Projection Collapse and Recursive Processing

Mathematical Theorem: If a representation factors through an observer map $\rho : X \rightarrow Y$, then any two states equivalent under ρ collapse to the same downstream toroidal representation. Downstream deterministic transformations cannot recover distinctions already lost at the observer interface.

Mathematical Theorem: If two states are already collapsed under projection, recursive processing (iteration of deterministic transformations on T^2) preserves that collapse. Recursion transforms internal structure but does not recover information absent from the interface.

11.4 Self Model Limitation

Mathematical Theorem: A self model map $\mu : S \rightarrow M$ that is many to one implies that distinct system states collapse into the same self model state. This is the safe version of the Godel self reference argument: incomplete self access follows from projection limits, not from applying Godel directly to brains.

11.5 Reconstructive Memory

Mathematical Theorem: A reconstruction map $\kappa : C \rightarrow R$ that is many to one implies that distinct cues can collapse into the same reconstructed state. Shared reconstruction errors can arise from shared reconstruction maps or attractor landscapes, not altered reality.

11.6 Invariance and Constrained Agency

Mathematical Theorem: If a transformation preserves an invariant, then structurally relevant content can survive changes in surface representation. This supports the claim that meaningful cognition requires preservation of relational invariants across representational transformations.

Mathematical Definition: Constrained agency is formalized as the existence of available actions under state dependent constraints, with outcome branching when allowed actions lead to different outcomes. This supports the constrained agency framing without claiming absolute free will.

11.7 Topology Evidence Boundaries

Mathematical Definition: A torus compatible Betti signature requires $\beta_0 = 1$, $\beta_1 = 2$, $\beta_2 = 1$. The Lean formalization verifies that H_1 evidence alone is not full torus evidence. Strong topological support requires H_1 , H_2 , persistence gap, controls, and methodological qualification.

12 Bounded Future Implications

The following subsections outline speculative extensions of the framework developed above. Each is labeled as future work or hypothesis and does not constitute a validated result of this paper. They are included to indicate the broader research program that the empirical and formal core is intended to support.

12.1 The Terahertz Measurement Gap

Future Work: Between the frequency range accessible through conventional electrophysiology and the molecular vibrational dynamics of neural tissue lies an observational gap spanning approximately 10 to 12 orders of magnitude. EEG and related techniques

measure activity up to roughly 10^3 Hz, while molecular and lattice oscillations operate at 10^9 to 10^{13} Hz. The intervening range remains largely unexplored in living neural tissue.

The terahertz band (~ 0.1 to 10 THz) corresponds to photon energies of approximately 0.4 to 41 meV, a scale that matches thermal fluctuations in biological tissue, weak intermolecular forces, and hydrogen bond energies. Fröhlich [15] proposed that biological systems supplied with constant metabolic energy can produce coherent longitudinal electric modes in this frequency region. Subsequent terahertz spectroscopy studies have confirmed that biological macromolecules exhibit collective vibrational modes in this range [16].

Hypothesis: The terahertz band is proposed here as a frequency domain hypothesis bridging molecular and cellular scales, not as an experimentally validated component of the present framework. No direct causal influence of terahertz radiation on grid cell dynamics or toroidal manifold formation is claimed. The hypothesis is that current neuroscience instrumentation was not designed to detect signals in this intermediate range, and that THz spectroscopy may eventually reveal collective electromagnetic dynamics in neural tissue relevant to the synchronization landscape of macroscopic oscillatory activity. Experimental validation will require dedicated terahertz spectroscopy measurements on living neural tissue.

12.2 Microbiome Evolutionary Pressure

Hypothesis: The human gastrointestinal tract harbors a microbial ecosystem of considerable complexity, comprising approximately 10^{13} to 10^{14} microorganisms [25]. The vagus nerve provides a direct neural pathway between the gut and the brainstem [26, 27]. The proposal advanced here is that the evolutionary pressure imposed by the need to manage a complex, fluctuating internal microbial ecosystem was a *contributing* driver of the neural pruning, hierarchical organization, and recursive internal modeling capacity that characterizes the mammalian brain. This is presented as a plausible evolutionary pressure rather than a single causal driver.

Organisms with better internal models of their gut ecosystem would have been able to extract more nutrients, resist more pathogens, and maintain more stable metabolic states. Over evolutionary time, this pressure would have favored nervous systems with greater capacity for hierarchical, recursive internal representation. The discrete frequency bands that characterize cortical oscillations [7] may reflect, in part, the need for a modular frequency regime capable of simultaneously representing information at multiple temporal scales.

Future Work: This hypothesis requires comparative neuroscience evidence across species with varying microbiome complexity, and does not constitute a validated finding of the present work.

12.3 Distributed Reconstruction and Shared Memory Errors

Future Work: Memory is not treated here as perfect local storage. The relevant question is whether memory depends on distributed, state dependent reconstruction constrained by neural, glial, bodily, and environmental context. Shared reconstruction errors are scientifically interesting not because they imply altered reality, but because they suggest that reconstruction landscapes may have common structure across observers. Future

work should investigate the role of semantic attractor landscapes, predictive processing, compression, shared priors, and possible glial/astrocytic long timescale stabilization in producing structured reconstruction errors.

12.4 Glial Contributions to Attractor Stabilization

Future Work: Given the role of astrocytes in synaptic regulation, calcium signaling, metabolic support, and network modulation, future work should ask whether glial architectures contribute to long timescale stabilization of neural attractor landscapes. This connects to neuronal geometric firing patterns, grid cell topology, astrocytic calcium waves, metabolic constraints, and slow timescale integration.

12.5 Invariance, Language, and Meaning Preservation

Future Work: Human experience depends on mappings that preserve enough relational structure across transformation, metaphor, compression, and context to remain meaningful. This connects to observer interface theory because perception, language, and memory all require invariant structure across changing representations. The general mapping principle is: meaningful cognition requires preservation of relational invariants across representational transformations. This motivates future formal work connecting topological invariance in neural manifolds to semantic invariance in linguistic and conceptual systems.

13 Conclusion

This paper has developed a physical dynamical framework explaining why toroidal manifolds arise in recursive neural systems, anchored by a methodological investigation of the limits of standard topological recovery pipelines.

We reanalyzed the Gardner et al. (2022) grid cell dataset, demonstrating that while standard PCA, UMAP, and Vietoris Rips pipelines can recover topological features compatible with toroidal organization, they are highly sensitive to dense temporal sampling and low variance embeddings. The lack of a clean persistence gap in the trajectory level analysis highlights the necessity of topology aware dimensionality reduction and denoising techniques for robust attractor reconstruction.

Building on this empirical anchor, we integrated the strongest formal material from observer interface theory: finite observation, non invertible projection, latent equivalence classes, entropy coupling, and recursive closure. We explicitly separated empirical claims from mathematical definitions, interpretations, and bounded speculative implications. This framework establishes a rigorous starting point for future theories of recursive self reference, constrained agency, and representational continuity, without requiring direct physical implementation of algebraic structures.

A Lean 4 Formalization

The following Lean 4 modules formalize the claim boundaries, mapping logic, and topology evidence rules used throughout this paper. The full verified code is available in the project repository.

A.1 TMRND__FormalSpine.lean

```

/-!
TMRND formal spine:
observer projection, observational equivalence,
and toroidal grid-cell phase mapping.
-/

variable {X 0 : Type}
variable (pi : X -> 0)

def ObsEq (pi : X -> 0) (x y : X) : Prop :=
  pi x = pi y

theorem ObsEq_refl (pi : X -> 0) (x : X) :
  ObsEq pi x x := by rfl

theorem ObsEq_symm (pi : X -> 0) (x y : X) :
  ObsEq pi x y -> ObsEq pi y x := by
  intro h; exact h.symm

theorem ObsEq_trans (pi : X -> 0) (x y z : X) :
  ObsEq pi x y -> ObsEq pi y z -> ObsEq pi x z := by
  intro hxy hyz; exact hxy.trans hyz

axiom S1 : Type
abbrev T2 := S1 x S1
abbrev T3 := S1 x S1 x S1

variable (phi1 phi2 : X -> S1)

def GridTorusMap (phi1 phi2 : X -> S1) (x : X) : T2 :=
  (phi1 x, phi2 x)

theorem same_phases_same_torus
  (phi1 phi2 : X -> S1) (x y : X)
  (h1 : phi1 x = phi1 y) (h2 : phi2 x = phi2 y) :
  GridTorusMap phi1 phi2 x = GridTorusMap phi1 phi2 y := by
  cases h1; cases h2; rfl

```

A.2 TMRND__TopologyEvidence.lean

```

/-!

```

Topology evidence rules:

Betti signatures, H1/H2 boundaries, and
reviewer-safe interpretation constraints.

-/

structure BettiSignature where

 beta0 : Nat

 beta1 : Nat

 beta2 : Nat

def torusCompatible (b : BettiSignature) : Prop :=

 b.beta0 = 1 && b.beta1 = 2 && b.beta2 = 1

inductive TopologyClaim

 | compatibleWith -- "features compatible with"

 | evidenceFor -- "evidence for"

 | suggestive -- "suggestive of"

-- H1 alone is NOT full torus evidence

def h1OnlyIsNotProof : Prop :=

 True -- axiomatically enforced as a manuscript rule

A.3 TMRND_ManuscriptRules.lean

/-!

Manuscript claim-type labels and boundary rules.

-/

inductive ClaimType

 | EmpiricalResult

 | MathematicalDefinition

 | MathematicalTheorem

 | Interpretation

 | Analogy

 | Hypothesis

 | FutureWork

-- Godel: analogy only

def godelBound : ClaimType := ClaimType.Analogy

-- Quaternions: mathematical definition only

def quaternionBound : ClaimType := ClaimType.MathematicalDefinition

-- Consciousness: hypothesis/future work only

def consciousnessBound : List ClaimType :=

 [ClaimType.Hypothesis, ClaimType.FutureWork]

-- Free will: interpretation only

```
def freeWillBound : ClaimType := ClaimType.Interpretation
```

References

- [1] Torkel Hafting, Marianne Fyhn, Sturla Molden, May-Britt Moser, and Edvard I. Moser. Microstructure of a spatial map in the entorhinal cortex. *Nature*, 436(7052): 801–806, 2005. doi: 10.1038/nature03721.
- [2] Hanne Stensola, Tor Stensola, Trygve Solstad, Kristian Frøland, May-Britt Moser, and Edvard I. Moser. The entorhinal grid map is discretized. *Nature*, 492(7427): 72–78, 2012. doi: 10.1038/nature11649.
- [3] KiJung Yoon, Michael A. Buice, Caswell Barry, Robin Hayman, Neil Burgess, and Ila R. Fiete. Specific evidence of low-dimensional continuous attractor dynamics in grid cells. *Nature Neuroscience*, 16(8):1077–1084, 2013. doi: 10.1038/nn.3450.
- [4] Richard J. Gardner, Erik Hermansen, Marius Pachitariu, Yoram Burak, Nils A. Baas, Benjamin A. Dunn, May-Britt Moser, and Edvard I. Moser. Toroidal topology of population activity in grid cells. *Nature*, 602(7895):123–128, 2022. doi: 10.1038/s41586-021-04268-7.
- [5] Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Westview Press, 2nd edition, 2015.
- [6] Henk W. Broer, George B. Huitema, and Mikhail B. Sevryuk. *Quasi-Periodic Motions in Families of Dynamical Systems: Order amidst Chaos*, volume 1645 of *Lecture Notes in Mathematics*. Springer, 1996. doi: 10.1007/BFb0093516.
- [7] György Buzsáki. *Rhythms of the Brain*. Oxford University Press, New York, 2006.
- [8] György Buzsáki and Andreas Draguhn. Neuronal oscillations in cortical networks. *Science*, 304(5679):1926–1929, 2004. doi: 10.1126/science.1099745.
- [9] Ryan T. Canolty, Erik Edwards, Sarang S. Dalal, Maryam Soltani, Srikantan S. Nagarajan, Heidi E. Kirsch, Mitchel S. Berger, Nicholas M. Barbaro, and Robert T. Knight. High gamma power is phase-locked to theta oscillations in human neocortex. *Science*, 313(5793):1626–1628, 2006. doi: 10.1126/science.1128115.
- [10] Adriano B. L. Tort, Robert W. Komorowski, Joseph R. Manns, Nancy J. Kopell, and Howard Eichenbaum. Theta-gamma coupling increases during the learning of item-context associations. *Proceedings of the National Academy of Sciences*, 106(49): 20942–20947, 2009. doi: 10.1073/pnas.0911331106.
- [11] John Lisman. The theta/gamma discrete phase code occurring during the hippocampal phase precession may be a more general brain coding scheme. *Hippocampus*, 15(7):913–922, 2005. doi: 10.1002/hipo.20121.
- [12] Si Wu, K. Y. Michael Wong, C. C. Alan Fung, Yuanyuan Mi, and Wenhao Zhang. Continuous attractor neural networks: candidate of a canonical model for neural information representation. *F1000Research*, 5:156, 2016. doi: 10.12688/f1000research.7387.1.
- [13] Erik Hermansen, David A. Klindt, and Benjamin A. Dunn. Uncovering 2-D toroidal representations in grid cell ensemble activity during 1-D behavior. *Nature Communications*, 15:5429, 2024. doi: 10.1038/s41467-024-49703-1.

- [14] Giordano Di Sarra, Siddharth Jha, and Yasser Roudi. The role of oscillations in grid cells' toroidal topology. *PLOS Computational Biology*, 21(1):e1012776, 2025. doi: 10.1371/journal.pcbi.1012776.
- [15] Herbert Fröhlich. Long-range coherence and energy storage in biological systems. *International Journal of Quantum Chemistry*, 2(5):641–649, 1968. doi: 10.1002/qua.560020505.
- [16] Andrea G. Markelz. Terahertz dielectric sensitivity to biomolecular structure and function. *IEEE Journal of Selected Topics in Quantum Electronics*, 14(1):180–190, 2008. doi: 10.1109/JSTQE.2007.913424.
- [17] Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, Berlin, 1984.
- [18] Steven H. Strogatz. From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D*, 143(1–4):1–20, 2000. doi: 10.1016/S0167-2789(00)00094-4.
- [19] Andrey N. Kolmogorov. On conservation of conditionally periodic motions under small perturbations of the Hamiltonian. *Doklady Akademii Nauk SSSR*, 98:527–530, 1954.
- [20] Vladimir I. Arnold. Proof of a theorem of A. N. Kolmogorov on the invariance of quasi-periodic motions under small perturbations of the Hamiltonian. *Russian Mathematical Surveys*, 18(5):9–36, 1963.
- [21] Jürgen Moser. On invariant curves of area-preserving mappings of an annulus. *Nachrichten der Akademie der Wissenschaften in Göttingen, Mathematisch-Physikalische Klasse*, pages 1–20, 1962.
- [22] Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010. doi: 10.1038/nrn2787.
- [23] Kurt Gödel. Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38(1):173–198, 1931.
- [24] Gregoire Nicolis and Ilya Prigogine. *Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order through Fluctuations*. Wiley-Interscience, New York, 1977.
- [25] John F. Cryan, Kenneth J. O’Riordan, Caitlin S. M. Cowan, Kiran V. Sandhu, Thomaz F. S. Bastiaanssen, Marcus Boehme, Martin G. Codagnone, Sofia Cusotto, Christine Fulling, Anna V. Golubeva, Katherine E. Guzzetta, Minal Jaggar, Caitriona M. Long-Smith, Joshua M. Lyte, Jason A. Martin, Alicia Molinero-Perez, Gerard Moloney, Emanuela Morelli, Enrique Morillas, Rory O’Connor, Joana S. Cruz-Pereira, Veronica L. Peterson, Kieran Rea, Nathaniel L. Ritz, Eoin Sherwin, Simon Spichak, Emily M. Teichman, Marcel van de Wouw, Ana Paula Ventura-Silva, Shauna E. Wallace-Fitzsimons, Niall Hyland, Gerard Clarke, and Timothy G. Dinan. The microbiota-gut-brain axis. *Physiological Reviews*, 99(4):1877–2013, 2019. doi: 10.1152/physrev.00018.2018.
- [26] Bruno Bonaz, Thomas Bazin, and Sonia Pellissier. The vagus nerve at the interface of the microbiota-gut-brain axis. *Frontiers in Neuroscience*, 12:49, 2018. doi: 10.3389/fnins.2018.00049.

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- [27] Christine Fülling, Timothy G. Dinan, and John F. Cryan. Gut microbe to brain signaling: what happens in vagus. . . . *Neuron*, 101(6):998–1002, 2019. doi: 10.1016/j.neuron.2019.02.008.